3.14 Principal Ideal Domains

Definition 3.14.1 (Principal Ideal Domain)
*An integral domain A is a **principal ideal domain** (or **PID**) if every ideal $\mathfrak{a}$ is principal.*

Proposition 3.14.2
A PID is Noetherian.

Proof. Suppose we have an ascending chain of ideals

$$\mathfrak{a}_1 \subseteq \mathfrak{a}_2 \subseteq \cdots \subseteq \mathfrak{a}_n \subseteq \cdots$$

Clearly the union is again an ideal, which is also principal of the form (a). We must have $a \in \mathfrak{a}_n$ for some n, whence it terminates after n. ∎

Proposition 3.14.3 (Integers form a PID)
$\mathbb{Z}$ is a PID.

Proof. This follows from the well-ordering principle. Let $\mathfrak{a}$ be an ideal with minimal positive element d. We claim $\mathfrak{a} = (d)$. By the division algorithm (or apply well-ordering principle to the coset $x + (d)$), for every $x \in \mathfrak{a}$ there is $0 \leq r < d$ and $q \in \mathbb{Z}$ such that

$$x = qd + r.$$

Clearly $r \in \mathfrak{a}$, whence by minimality $r = 0$ as required. ∎

Proposition 3.14.4
Let k be a field then the polynomial ring $k[X]$ is a PID

Proof. Let $\mathfrak{a} \triangleleft k[X]$ be an ideal. Choose $f \in \mathfrak{a}$ to have minimal degree, then we claim $\mathfrak{a} = (f)$. For $g \in \mathfrak{a}$ we have by (3.7.14) $g = qf + r$ for $\deg(r) < \deg(f)$. Clearly $r \in \mathfrak{a}$, so by minimality $r = 0$ and the result follows. ∎

Lemma 3.14.5 (Co-prime elements in a PID)
Let A be a PID, then x, y are coprime if and only if they have no non-invertible common divisors.

Proof. First suppose $(x, y) = A$, then $ax + by = 1$ and any common divisor d must divide 1 and therefore be invertible.

Conversely suppose $(x, y) \neq (1)$, since A is a PID it must equal (d) for some non-invertible d which is then a common divisor. ∎

Proposition 3.14.6
Let A be a principal ideal domain, M a free A-module with rank m and $N \subset M$ an A-submodule. Then N is free with rank $n \leq m$.

Proof. The rank of a finite free A-module is well-defined (3.8.15).

We proceed by induction on m. If $m = 0$ then $M = 0$ and so $N = 0$ is free of rank 0. Suppose $m \geq 1$ and the result holds for rank $m - 1$. Let $x_1, ..., x_m$ be a basis for M and let $\pi : M \to A$ be the map $\sum_i c_i x_i \mapsto c_m$. Its kernel is the free submodule M' generated by $x_1, ..., x_{m-1}$, of rank $m - 1$. Then $N' := N \cap M'$ is the kernel of $\pi|_N$ and a submodule of M', hence free of rank $n' \leq m - 1$ by the induction hypothesis. The image $\pi(N)$ is an ideal of A, hence equal to (a) for some $a \in A$.

If $a = 0$ then $N = N'$ and we are done. Suppose $a \neq 0$ and choose $y \in N$ with $\pi(y) = a$. We claim $N = N' \oplus Ay$ with $Ay \cong A$. For $z \in N$ we have $\pi(z) = ca$ for some $c \in A$, whence $z - cy \in N'$. Moreover $\pi(cy) = ca$ is non-zero for $c \neq 0$, as A is an integral domain. Hence $c \mapsto cy$ is injective, so $Ay \cong A$, and $N' \cap Ay = 0$ as $cy \in N'$ forces $ca = 0$ and so $c = 0$. Therefore N is free with basis a basis of N' together with y, of rank $n' + 1 \leq m$. ∎

Proposition 3.14.7
Let A be a principal ideal domain and M a finitely generated, torsion-free A-module. Then M is a finite free A-module.

Proof. Let $\{m_1, ..., m_n\}$ be a generating set for M, and $\{v_1, ..., v_r\}$ an arbitrary linearly independent subset of M. Then by definition we have

$$v_i = \sum_{j=1}^{n} a_{ij} m_j \quad a_{ij} \in A$$

Consider the elements of A^n given by

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$$w_i := (a_{i1}, ..., a_{in})$$

Then we can see that $\{w_1, ..., w_r\}$ is an A-linearly independent subset of A^n, and therefore a K-linearly independent subset of K^n for $K = \mathrm{Frac}(A)$ (a K-linear dependence gives an A-linear dependence on clearing denominators). As K^n has the standard basis of order n, it has dimension n over K, so $r \leq n$ (3.3.126).

Therefore we may choose $\{v_1, ..., v_r\}$ to be a maximal linearly independent subset and define $M' := \langle v_1, ..., v_r \rangle$. For each generator m_i, by maximality of the subset there must exist an $a_i \neq 0$ such that

$$a_i m_i \in M'$$

Define $a := a_1 ... a_n$ then this satisfies $aM \subseteq M'$, since the m_i generate M. As A is an integral domain we have a series of injections

$$M \hookrightarrow aM \subseteq M' \xrightarrow{\sim} A^r$$

Therefore we are done by (3.14.6). ■

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